Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/30/2005
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Project Title:	High LET Radiation And Neurogenesis: Implications And Mechanisms Underlying Cognitive Impairment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-02
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No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
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	For those engaged in space travel, the radiation environment poses a range of potential health risks. For instance, it is possible that exposure to high-energy protons and heavy ions within galactic cosmic rays or solar particle events can induce a range of CNS injuries, including cognitive impairment. While the pathogenesis of radiation-induced cognitive impairment is not clear, we have shown that neuronal precursor cells in the hippocampus may be involved. Using low doses of X-rays (1-5 Gy), we have demonstrated, in vitro and in vivo, that neural stem cells of the hippocampus are exquisitely sensitive to irradiation. In addition, these cells exhibit oxidative stress and undergo apoptosis after irradiation. Low dose x-irradiation of these cells in vitro also leads to a persistent oxidative stress that coincides temporally with a dose-dependent reduction in neurogenesis observed in vivo. Consequently, we hypothesize that oxidative stress plays a contributory if not causative role in regulating the radiation response of neural stem cells in vitro and in vivo. Further, we		

Task Description:	suggest that oxidative stress constitutes a biochemical mechanism that can influence the fate of stem cell differentiation in vitro and hippocampal neurogenesis in vivo. It is our overall objective to determine how oxidative stress affects the high LET radiation response of precursor cells in the subgranular zone (SGZ) of the hippocampus and to identify compounds/strategies that will enable us to ameliorate the adverse effects of high LET irradiation on hippocampal neurogenesis. We have defined 4 specific aims to address our objective. Aim 1: Using in vitro and in vivo models, determine if deficiencies in site-specific superoxide dismutases (SODs) lead to acute (days) oxidative stress and altered neurogenesis after exposure to high-energy protons. Aim 2: Using in vitro and in vivo models, determine if deficiencies in site-specific superoxide dismutases (SODs) lead to chronic (months) oxidative stress and altered differentiation/neurogenesis following exposure to high-energy iron ions. Aim 3: Determine if antioxidant treatment with SOD/catalase mimetic agents reduce high-LET induced radiation injury to neural stem cells in vitro and ameliorate the radiation-induced inhibition of neurogenesis in vivo. Aim 4: Quantify the effects of high-energy iron ions on cognitive function in wild type mice and determine if antioxidant treatment ameliorates those effects. Determine if or how oxidative stress impacts the radioresponse of neural precursor cell function. Elucidating oxidative stress as a biochemical mechanism regulating the behavior of neural precursor cells will provide significant insight into how irradiation might lead to impaired cognition during spaceflight. These studies will also provide information relevant to the development of antioxidant strategies to ameliorate the adverse effects of irradiation in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work will help define the risks associated with radiation exposure in space. By understanding these risks NASA can improve its strategies for promoting the health of astronauts.
Task Progress:	Progress over our second funding period has been significant. We have undertaken our first BNL run (NSRL-5), and our second trip to Loma Linda where we successfully irradiated a large number of in vitro samples as well as our SOD3 deficient mice. The start of our animal studies marks a significant milestone. These experiments utilized 24 controls (WT), 24 ECSOD-/+ (heterozygotes) and 36 ECSOD-/- (homozygous knockout) mice subjected to 0, 0.5 or 2 Gy of 56Fe ions. These mice are currently being processed for the analysis of neurogenesis and we expect to have results by our next progress report. We anticipate that these studies will corroborate our X-ray studies and show that under conditions of increased oxidative stress, neurogenesis is inhibited. Major accomplishments are listed below: 1) In vitro studies using hippocampal and subventricular neural precursor cells have revealed that 56Fe-ions induce marked levels of oxidative stress. Reactive oxygen species detected after heavy ion irradiation are 2-10 fold higher at various times (6-48h) after irradiation. 2) In vitro studies using hippocampal neural precursor cells have shown protons to be more efficient at eliciting oxidative stress than X-rays. 3) The antioxidant lipoic acid has shown promise at reducing radiation-induced ROS, whether added as a radioprotector (before irradiation) or as a mitigating agent (after irradiation). 4) Mice deficient in SOD3 (extracellular SOD) show reduced neurogenesis, providing compelling evidence that oxidative stress is a biochemical modifier of neurogenesis. 5) Funding from this Grant has resulted in 5 published manuscripts (see bibliography) In summary we have determined that neural precursor cells appear uniquely predisposed to redox sensitive changes that are likely to have a significant impact on the radioresponse of these cells in vitro and in vivo. These cells show significant and dose-dependent increases in ROS that may well underlie the prolonged inhibition of neurogenesis following irradiation. Our work provides ample evidence that oxidative stress is a biochemical mechanism that regulates neurogenesis, and that interventions aimed at minimizing oxidative stress in the CNS have the potential to ameliorate the adverse impact of the space radiation environment.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Rola, R.; Otsuka, S.; Obenaus, A.; Nelson, G.A.; Limoli, C.L.; VandenBerg, S.R.; Fike, J. "Indicators of hippocampal neurogenesis are altered by 56Fe irradiation in a dose dependent manner" Radiat Res. 162:442-446 , Oct-2004
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